

Getting K-Combinations with Recursion

Recall that order is not significant for combinations in the way it is for permutations. Yet generating all k -combinations of a set is a bit tricky because you don't want your algorithm to generate duplicates: if you create the AB 2-combination from the set {A, B, C}, you don't want to also create BA, because it's the same 2-combination as AB.

To figure out how we can write recursive code to solve this problem, let's see how a tree can visually describe generating all the k -combinations of a set. [Figure 6-4](#) shows a tree with all the combinations from the set {A, B, C, D}.

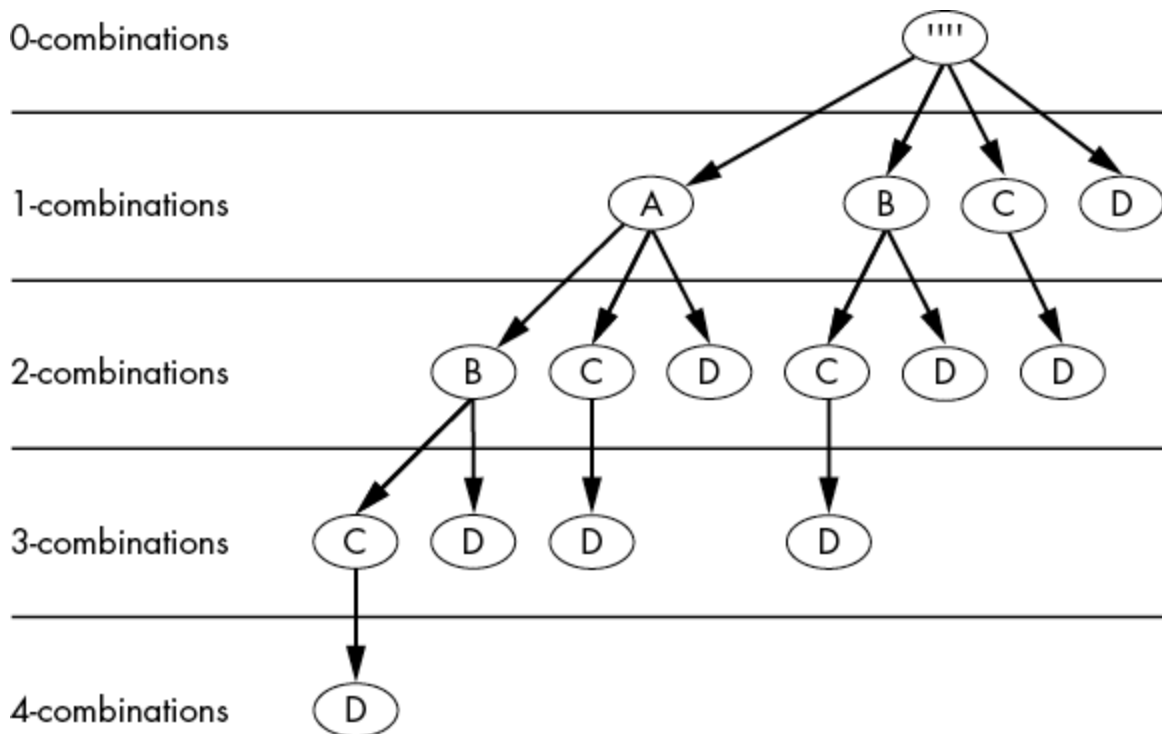


Figure 6-4: Tree showing every possible k -combination (from 0 to 4) from the set {A, B, C, D}

To gather, for example, 3-combinations from this tree, start at the root node at the top and do a depth-first tree traversal to the 3-combinations level, while memorizing each node's letter on the way to the bottom. (Depth-first searches are discussed in Chapter 4.) Our first 3-combination